

Suad Arzimanli

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ACADEMIC AND RESEARCH PROFILE

Geospatial systems practitioner and researcher with a background in Mathematics and Computer Science and applied experience in enterprise GIS, spatial data engineering, cartography, and geospatial software development. Research interests include geospatial intelligence (GEOINT), web-based GIS, spatial analysis, and the design of operational geospatial systems that connect data preparation, analysis, visualization, and decision support.

Combines hands-on work in ArcGIS (Pro, Online, Experience Builder), Python/ArcPy automation, enterprise geodatabases, and cartographic production with full-stack software engineering (C#/.NET, SQL, REST APIs) for building and maintaining production geospatial applications in municipal and infrastructure contexts. Experienced in data validation and QA/QC, map product dissemination (reports, briefings, training), and technical leadership of GIS-oriented delivery teams.

EDUCATION

Master of Science in Mathematics and Computer Science

2025 – Present

University of Łódź, Poland

Program combines mathematical and computational foundations applicable to modeling, data analysis, and geospatial systems.

Bachelor of Science in Mathematics and Computer Science

Oct 2021 – July 2025

University of Łódź, Poland

Thesis: “Application using Geospatial Intelligence (GEOINT) to map and analyze geographic data” — web-based GIS system for conflict prediction and asset management. Full-time studies in English.

RESEARCH AND GEOSPATIAL PROJECTS

Bachelor thesis — GEOINT web GIS (University of Łódź)

Designed and implemented a web-based GIS application applying geospatial intelligence concepts to map and analyze geographic data, with emphasis on spatial visualization and geographic analysis for conflict prediction and asset management contexts.

Zürich Tree Management

Municipal tree governance demonstrator: service-request workflow from map intake through inspection to closure; City of Zürich open Baumkataster (WFS); ASP.NET Core; bilingual UI.

GEOINT Crisis Zones

Interactive mapping of 7,794 ACLED conflict and protest events in Poland (2020–2024) with filters, charts, and evaluated city-level forecasts; ASP.NET Core 8, Google Maps, Chart.js.

Water Leak Monitoring Dashboard

Infrastructure GIS demonstrator: synthetic SCADA readings, zone-level risk scoring, persisted operator incident from alert through close-out.

Urban Heat Island — Łódź

Sentinel-2 NDVI and Landsat-8 LST on a 500 m grid (1,544 cells); regression, classified rasters, Leaflet web map; Google Earth Engine pipeline.

GeoLineage

File-based lineage receipts for STAC → GeoParquet pipelines; minimum recompute, diff-since-last-run views, Streamlit inspection UI; Python.

PROFESSIONAL EXPERIENCE

Team Lead GIS & Full-Stack .NET Developer

Dec 2023 – Present

DZ CIS LLC — Remote

Intern — training and foundation (Dec 2023 – Feb 2024)

- Joined in a trainee/intern capacity with hands-on training in Cityworks AMS, enterprise GIS workflows, ArcGIS Pro, and geodatabase structures.
- Assisted senior developers with backend (.NET, SQL) and frontend (map-based web interfaces) tasks for asset management and service-request applications.
- Gained practical experience with geospatial datasets, municipal service-request lifecycles, and infrastructure-oriented operational data.

Junior full-stack developer (Mar 2024 – Nov 2024)

- Designed, built, and maintained enterprise geodatabases; ensured schema consistency, data integrity, and operational readiness.
- Performed geospatial analysis, spatial queries, data cleaning, validation, and QA/QC using SQL and Python (null attributes, duplicates, geometry and projection consistency).
- Processed GIS data from Excel, CSV, GeoJSON, SQL, and geodatabase sources; prepared map packages and pre-publish validation inputs for ArcGIS MapServer/FeatureServer deployment workflows.
- Integrated RESTful services among Cityworks AMS, Siebel 311, ArcGIS Feature Services, and internal databases; consumed services via direct ArcGIS REST and .NET API layers.
- Developed full-stack geospatial applications (Google Maps API, .NET MVC, ArcGIS services) for Tree Management, pothole/311-style, and DOT/highway-related programs.
- Engineered interactive map tools: custom markers, clustering, radius filtering, overlays, location search, and consistent multi-filter behavior.
- Built ArcGIS Experience Builder applications and custom widgets; maintained ArcGIS Online web maps and layers.
- Built and maintained Windows Forms utilities and custom internal tools supporting data and operational workflows.
- Produced recurring ArcGIS Pro cartographic outputs (layouts, print/PDF, template-driven map series) for asset-focused reporting and briefings.
- Automated geoprocessing with Python/ArcPy, including custom flows where built-in ArcGIS tools did not cover full workflow requirements; applied raster and Spatial Analyst operations as needed.

- Developed modular C#/.NET backend components with asynchronous patterns and structured controller design.
- Produced technical documentation, workflow diagrams, and user guides; delivered training, demos, and briefings on maps and spatial analysis.

Senior full-stack developer (Dec 2024 – Nov 2025)

- Upgraded and modernized GIS-backed .NET applications; migrated framework versions and maintained release discipline.
- Led delivery of ASP.NET Core MVC/Razor applications, coordinating JavaScript map UI behavior with API and service-layer contracts.
- Managed IIS-hosted deployments and environment-specific configuration for integrated geospatial and enterprise applications.
- Resolved production authentication and integration issues (Entra ID/OIDC, cookie/session middleware, API gateway contracts, reverse-proxy headers, 401/403 handling for AJAX clients).
- Aligned request/response contracts across frontend payloads, controller DTOs, service mappings, and SQL persistence for multi-step operational workflows.
- Executed safe schema and configuration evolution (additive database changes, backward-compatible config cleanup) without disrupting production GIS services.

Team lead (Dec 2025 – Present)

- Technical leadership across GIS and .NET projects: task planning, coordination, mentoring, and engineering standards.
- Led a dedicated team building ArcGIS Experience Builder applications for custom operational systems.
- Designed and implemented an internal team management application for task tracking and coordination.
- Drove release reliability through smoke and regression checks across core map and application workflows.
- Led onboarding and training; presented map products and spatial analysis to technical and non-technical stakeholders.
- Delivered work in Agile/Scrum with a remote, cross-functional team under Time and Materials engagement.

TECHNICAL COMPETENCIES

GIS platforms, spatial analysis, and cartography

- ArcGIS Pro, ArcMap, ArcGIS Online, ArcGIS Experience Builder; custom Experience Builder widgets
- Enterprise geodatabases, spatial and non-spatial SQL, projection and coordinate-system awareness
- GIS data pipeline: Excel/CSV/GeoJSON/SQL to layers/geodatabases; map package preparation for MapServer/FeatureServer workflows
- Cartography: layouts, print/PDF exports, template-driven map series, symbology and labeling for diverse audiences
- Raster geoprocessing and Spatial Analyst tools; broad ArcGIS Pro geoprocessing toolchains
- Python and ArcPy for geoprocessing automation, data cleaning, QA/QC, and custom workflow

scripts

- Google Maps API (markers, clustering, overlays, geofencing, search) in web mapping contexts

Geospatial data quality and dissemination

- QA/QC: null attributes, duplicates, geometry validity, projection consistency
- Training sessions, demos, presentations, and technical writing for map products and spatial analysis

Asset management and municipal GIS integrations

- Cityworks AMS (workflow automation, inspections, service requests)
- 311 and Siebel integrations; Tree Management and DOT/highway asset programs
- REST API integration between GIS services, asset platforms, and enterprise databases

Software engineering (geospatial systems context)

- C#, .NET 6/7/8, ASP.NET Core MVC, ASP.NET Core Web API, Windows Forms, Entity Framework Core
- JavaScript, jQuery, HTML/CSS, Razor; server-rendered web applications with complex UI state
- SQL Server, relational modeling, safe schema evolution
- IIS, SSO/Microsoft Entra ID, secure APIs, API management practices, Azure-aware deployment
- Git, Azure DevOps exposure, unit/integration testing when required; Agile/Scrum delivery
- Visual Studio, Postman, Microsoft Office

LANGUAGES

Azerbaijani — Native	English — Professional (C1/C2)
Turkish — Native	German — Intermediate (A2–B1)